MA 17: HOW TO SOLVE IT
PROBLEM SET 3

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This set has ten problems, arranged in roughly increasing order of difficulty. Any problem may count as one of your three submitted problems.

Due: 11:59PM of October 30. Late submissions will not be accepted.

Instructions:

- *Please submit solutions to at most 3 problems, written carefully. You are encouraged to work on more, but the TA will provide feedback on only 3. I recommend selecting the 3 for which you most want feedback.*
- *Collaboration on homework before you have solved at least 3 problems independently is discouraged, though not strictly forbidden. You should make a serious effort to work through each problem independently before turning to classmates, using the internet, or asking AI.*
- *As long as your submission clearly shows attempts on at least 3 problems, you will earn 8 points toward the final grade. The correctness of your solutions will not affect your points, although feedback will be provided.*
- *Submission should be made through Canvas.*

Problem 1.
Suppose $a < b$ and $f : [a, b] \to \mathbb{R}$ is continuous, with a derivative at every point of (a, b). Let $M = (\alpha, \beta)$ lie on the line through $(a, f(a))$ and $(b, f(b))$, where $\alpha \notin [a, b]$. Prove that for some $c \in (a, b)$, the tangent line to the graph of f at $(c, f(c))$ passes through M.

Problem 2. (Putnam 2020, A1; Handout 2, Problem 9)
How many positive integers N satisfy all of the following three conditions?

(i) N is divisible by 2020.
(ii) N has at most 2020 decimal digits.
(iii) The decimal digits of N are a string of consecutive ones followed by a string of consecutive zeros.

Problem 3. (Putnam 2022, A1)
Determine all ordered pairs of real numbers (a, b) such that the line $y = ax + b$ intersects the curve $y = \ln(1 + x^2)$ in exactly one point.

Problem 4. (Putnam 2022, B1)
Suppose that $P(x) = a_1x + a_2x^2 + \cdots + a_nx^n$ is a polynomial with integer coefficients, with a_1 odd. Suppose that $e^{P(x)} = b_0 + b_1x + b_2x^2 + \cdots$ for all x. Prove that b_k is nonzero for all $k \geq 0$.

Problem 5. (Putnam 2019, B2)
For all $n \geq 1$, let

$$a_n = \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right)\cos^2\left(\frac{k\pi}{2n}\right)}.$$

Determine

$$\lim_{n\to\infty} \frac{a_n}{n^3}.$$

Problem 6. (Putnam 2025, A2)
Find the largest real number a and the smallest real number b such that

$$ax(\pi - x) \leq \sin x \leq bx(\pi - x)$$

for all x in the interval $[0, \pi]$.

Problem 7. (Putnam 2024, B3)
Let r_n be the nth smallest positive solution to $\tan x = x$, where the argument of tangent is in radians. Prove that

$$0 < r_{n+1} - r_n - \pi < \frac{1}{(n^2+n)\pi}$$

for $n \geq 1$.

Problem 8.
Let $0 \leq a_0 < a_1 < a_2 < \cdots$ be integers. Suppose every nonnegative integer m has exactly one representation

$$m = a_i + 2a_j + 4a_k,$$

where $i, j, k \geq 0$ and the indices need not be distinct. Determine a_{1998}.

Problem 9. (Putnam 1990, B5)
Is there an infinite sequence $a_0, a_1, a_2, \ldots$ of nonzero real numbers such that for $n = 1, 2, 3, \ldots$ the polynomial

$$p_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

has exactly n distinct real roots?

Problem 10. (Putnam 2023, A3)
Determine the smallest positive real number r such that there exist differentiable functions $f\colon \mathbb{R} \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$ satisfying

(a) $f(0) > 0$,
(b) $g(0) = 0$,
(c) $|f'(x)| \leq |g(x)|$ for all x,
(d) $|g'(x)| \leq |f(x)|$ for all x, and
(e) $f(r) = 0$.